Summary in Graph

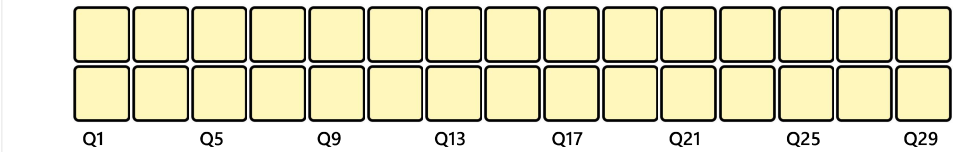
Exam Summary (GO Classes DA Test Series 2026 | Calculus and Optimisation | Subject Wise Test 1)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**30** (15 + 15)
Total Marks:	**45** (15 + 30)
Exam Duration:	**90 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (37) out of 368 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Suppose that f is continuous on $[a, b]$ and differentiable on (a, b). If $f'(x) > 0$ on (a, b). Which of the following is necessarily true?

A. f is decreasing on $[a, b]$,
B. f has no local extrema on (a, b),

C. f is a constant function on (a, b),
D. f is concave up on (a, b),

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function f has the property that $f(3) = 2$ and $f'(3) = 4$. Using a linear approximation to f near $x = 3$, an approximation to $f(2.9)$ is

A. 1.4
B. 1.6
C. 1.8
D. 1.9

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

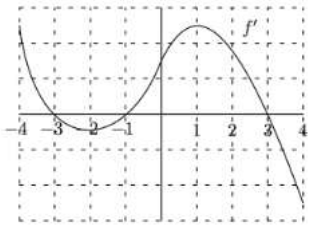
Which of the following are true about the function f if its derivative is $f'(x) = (x-1)^2(4-x)$?

I. f is decreasing for all $x < 4$.
II. f has a local maximum at $x = 1$.
III. f is concave up for all $1 < x < 3$.

A. I only
B. II only
C. III only
D. II and III only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus



The graph of the derivative of a function f is shown above. Which of the following are true about the original function f ?

I. f is increasing on the interval $(-2, 1)$.
II. f is continuous at $x = 0$.
III. f has an inflection point at $x = -2$.

A. I only
B. II only
C. III only
D. II and III only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Given that the function f is continuous at the point ($c, f(c)$), determine which of the following statements could be false.

A. $\lim_{x \to c} f(x)$ exists.
B. $\lim_{x \to c} f(x) = f(c)$.
C. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$.
D. $\lim_{x \to c} f(x) = c$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If $f(x) = 2 + |x - 3|$ for all x, then the value of the derivative $f'(x)$ at $x = 3$ is

A. -1
B. 0
C. 1
D. nonexistent

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f : [a, b] \to \mathbb{R}$ be a differentiable function, and define $g(x) = (f(x))^2$.
Apply the Mean Value Theorem (MVT) to the function g on the interval $[a, b]$.
Then which of the following is correct?

A. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = f'(c)^2$$

B. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = 2f(c)f'(c)$$

C. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = f(c)^2$$

D. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b-a} = f'(c)[f(a) + f(b)]$$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

$\lim_{n\to\infty} \frac{2^{n+1}+3^{n+1}}{2^n+3^n}$ equals

A. 3

B. 2

C. 1

D. 0

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let

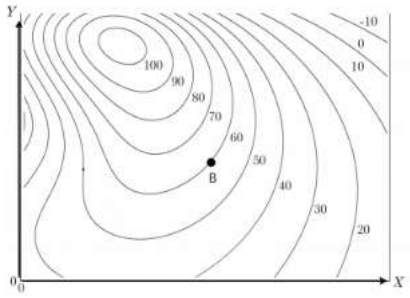
$$f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)$$

The number of distinct real roots of the equation $\frac{d}{dx} f(x) = 0$ is exactly

A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus



Based on the given contour plot of the function $h(x, y)$, determine the correct signs of the partial derivatives at point B :
What is the sign of $\frac{\partial h}{\partial x}$ and $\frac{\partial h}{\partial y}$ at point B ?

A. $\frac{\partial h}{\partial x} > 0, \quad \frac{\partial h}{\partial y} > 0$

B. $\frac{\partial h}{\partial x} < 0, \quad \frac{\partial h}{\partial y} > 0$

C. $\frac{\partial h}{\partial x} < 0, \quad \frac{\partial h}{\partial y} < 0$

D. $\frac{\partial h}{\partial x} = 0, \quad \frac{\partial h}{\partial y} > 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the following two statements about a function $f(x, y)$:

Statement I:

There exists a function $z = f(x, y)$ such that

$$f_{xx}(x, y) = 0, \quad f_{yy}(x, y) = 0, \quad \text{and} \quad f_{xy}(x, y) \neq 0$$

Statement II:

If $f_x(P_0) = 0$ and $f_y(P_0) = 0$, then P_0 is a local maximum or a local minimum of f.

Which of the following is correct?

A. Both Statement I and Statement II are true
B. Statement I is true, but Statement II is false
C. Statement I is false, but Statement II is true
D. Both Statement I and Statement II are false

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f(x) = x^3 + x^2$, and suppose gradient descent starts at $x_0 = -1$. What will be the direction of the first update step?
For all options: "Left" means towards more negative values of x, and "Right" means towards more positive values of x.

A. Move Right, because the slope is negative

B. Move Left, because the slope is positive

C. Move Right, because the function value is decreasing

D. Stay at the same point, since it's a critical point

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function. Consider the following statements regarding continuity and differentiability.
Which of the following statements is/are CORRECT?

A. Every function f is continuous.
B. Every continuous function is differentiable.
C. Every differentiable function is continuous.
D. There exist continuous functions which are not differentiable.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The positive integers m and n are relatively prime, and chosen in such a way that

$$\lim_{x\to 2} \frac{5\sqrt{1+4x} - 3\sqrt{1+12x}}{x-2} = -\frac{m}{n}.$$

What is $n - 4m$?

A. -1
B. 2
C. -3
D. 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

$$f(x) = \begin{cases} 2x-2 & \text{for } x < 3 \\ 2x-4 & \text{for } x \geq 3 \end{cases}$$

Let f be the piecewise-linear function defined above. Which of the following statements are true?

I. $\lim_{h\to 0^-} \frac{f(3+h)-f(3)}{h} = 2$

II. $\lim_{h\to 0^+} \frac{f(3+h)-f(3)}{h} = 2$

III. $f'(3) = 2$

A. None
B. II only
C. I and II only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Which of the following statements is always true for a function $f(x)$?

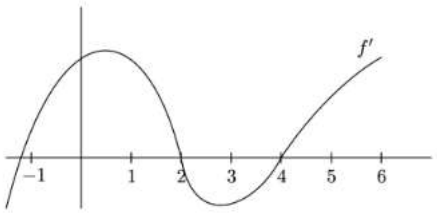
i. If $f(x)$ and $g(x)$ are continuous at $x = a$, then $\frac{f(x)}{g(x)}$ is continuous at $x = a$.
ii. If $f(x) + g(x)$ is continuous at $x = a$ and $f'(a) = 0$, then $g(x)$ is continuous ta $x = a$.

iii. If $f(x)+g(x)$ is differentiable at $x=a$, then $f(x)$ and $g(x)$ are both differentiable at $x=a$

A. only i.
B. only ii.
C. only iii.
D. i. and ii.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



The graph of the first derivative f' for a function f is shown above. Which of the following statements must be true?

I. $f(4)=f(-1)$
II. $f(2)>f(4)$
III. $f''(2)<f''(4)$

A. I only
B. II only
C. II and III only
D. I and III only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If f is continuous at $x=2$, and if f is defined as

$$f(x)=\begin{cases}\frac{\ln x^2-x\ln x}{x-2} & x\neq 2\\ k & x=2,\end{cases}$$

then k is

A. -1
B. -2
C. $-\frac{e}{4}$
D. $-\ln 2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f'(x)$ and $g'(x)$ exist and $f'(x) > g'(x)$ for all real x, then the graph of $y = f(x)$ and the graph of $y = g(x)$

A. intersect exactly once.
B. intersect no more than once.
C. do not intersect.
D. could intersect more than once.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $a > 1$ and $f, g, h : [-a, a] \longrightarrow \boldsymbol{R}$ be twice differentiable functions such that for some c with $0 < c < 1 < a$,

$$f(x) = 0 \text{ only for } x = -a, 0, a$$
$$f'(x) = 0 = g(x) \text{ only } x = -1, 0, 1$$
$$g'(x) = 0 = h(x) \text{ only for } x = -c, c$$

The possible relations between f, g, h are

A. $f = g'$ and $h = f'$

B. $f' = g$ and $g' = h$

C. $f = -g'$ and $h' = g$

D. $f = -g'$ and $h' = f$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

For every function $f : [0, 1] \to \mathbb{R}$ which is twice differentiable and satisfies $f'(x) \geq 1$ for all $x \in [0, 1]$, we must have

A. $f''(x) \geq 0$ for all $x \in [0, 1]$

B. $f(x) \geq x$ for all $x \in [0, 1]$

C. $f(x_2) - x_2 \leq f(x_1) - x_1$ for all $x_1, x_2 \in [0, 1]$ with $x_2 \geq x_1$

D. $f(x_2) - x_2 \geq f(x_1) - x_1$ for all $x_1, x_2 \in [0, 1]$ with $x_2 \geq x_1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

$\lim_{n\to\infty} \frac{1-2+3-4+5-6+\ldots+(-2n)}{\sqrt{n^2+1}+\sqrt{n^2-1}}$ equals

A. ∞
B. $1/2$
C. 0
D. $-1/2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f(x,y) = \frac{ax^2+by^2}{xy}$, where a and b are constants. If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $x = 1$ and $y = 2$, then the relation between a and b is:

A. $a = \frac{b}{4}$

B. $a = \frac{b}{2}$

C. $a = 2b$

D. $a = 4b$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let

$$f(x,y) = 15 - x^2 + 2y^2 + 6x - 8y.$$

Which of the following statements is correct about the critical point(s) of $f(x,y)$?

A. The function has a critical point at $(3,2)$, which is a local maximum.
B. The function has a critical point at $(3,2)$, which is a local minimum.
C. The function has no critical points.
D. The function has a critical point at $(3,2)$, which is a saddle point.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the function $f(x) = \sqrt{3-x}$. Which of the following is the definition of $f'(-1)$?

A. $f'(-1) = \lim_{x\to -1} \frac{\sqrt{3-x}-2}{x+1}$

B. $f'(-1) = \lim_{x\to \infty} \frac{\sqrt{3-x}-2}{x+1}$

C. $f'(-1) = \lim_{h\to 0} \frac{\sqrt{3-x-h}-\sqrt{3-x}}{h}$

D. $f'(-1) = \lim_{h\to 0} \frac{h}{\sqrt{3-x-h}-\sqrt{3-x}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

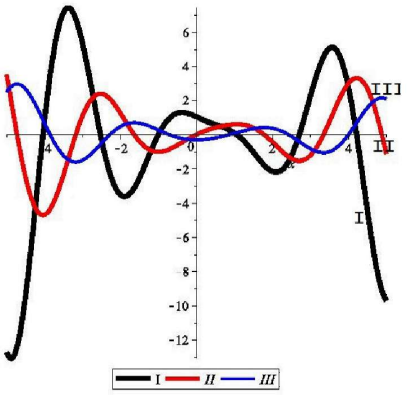
Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Calculus

The function $G(x) = (x^2 + m)e^x$, defined for all real values of x and m a fixed real constant, has two distinct inflection points if and only if $m <$ ___.

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



In the accompanying figure we have the graphs of f, f' and f''.
Identify these graphs with the roman numerals shown.

A. $I = f$
$II = f'$

B. $I = f$

C. $II = f$
$III = f'$

D. $III = f$
$II = f'$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The function $f(x) = x^2e^x$ is convex downward on the interval $[a, b]$, convex upward on $(-\infty, a]$ and also on $[b, \infty)$. Then what is ab ?

A. 1

B. 2
C. 3
D. 4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The function f is defined on the closed interval $[2, 4]$ and $f(2) = f(3) = f(4)$. On the open interval $(2, 4)$, f is continuous and strictly decreasing. Which of the following statements is true

A. f attains neither a minimum value nor a maximum value on the closed interval $[2, 4]$.
B. f attains a minimum value but does not attain a maximum value on the closed interval $[2, 4]$.
C. f attains a maximum value but does not attain a minimum value on the closed interval $[2, 4]$.
D. f attains both a minimum value and a maximum value on the closed interval $[2, 4]$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

x	0	1	2
$f(x)$	1	k	2

The function f is continuous on the closed interval $[0, 2]$ and has values that are given in the table above. The equation $f(x) = \frac{1}{2}$ must have at least two solutions in the interval $[0, 2]$ if $k =$

A. 0

B. $\frac{1}{2}$

C. 1

D. 2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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